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Remaining useful life prediction method for rolling bearings based on Mamba-SDP

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Abstract Rolling bearings are crucial components in industrial systems, and their health status is vital for system performance and stability. To address the limitations of traditional models in extracting spatiotemporal features and processing multisensor data, this paper proposes a remaining useful life (RUL) prediction method for rolling bearings using the Mamba-SDP framework. First, a channel fusion strategy efficiently integrates multisensor data to enhance feature representation under complex operating conditions. Second, the Mamba module extracts spatiotemporal deep features, while the improved channel fast Fourier transform (ICFFT) module processes non-stationary signals for improved frequency analysis. In addition, a scaled dot product (SDP) attention mechanism combined with cross-normalization mitigates numerical fluctuations in high-dimensional spaces, improving model stability and computational efficiency in capturing local and global dependencies. Furthermore, residual connections integrate SDP-weighted shallow features with deep features extracted by Mamba and ICFFT, maximizing the exploitation of both prior and posterior information. Finally, the fused features are fed into fully connected layers for RUL prediction. Extensive experiments on PHM 2012 and XJTU-SY datasets demonstrate that our method achieves MAE, RMSE, and score of 8.99, 11.50, 0.82 and 9.20, 11.40, 0.81, respectively, outperforming several state-of-the-art benchmarks with up to 7.1 % and 8.3 % lower MAE and RMSE, respectively, and a 4 % improvement in score. These results validate the effectiveness and generalization of the proposed approach, providing a promising solution for practical RUL prediction and industrial health management.

1. Introduction

Prognostics and health management (PHM) is a key technology that integrates health management, predictive modeling, and data analysis. By monitoring, diagnosing, and predicting equipment's condition in real time, PHM can detect potential issues early and take effective actions, thereby reducing unplanned downtime and extending the equipment's life span [1-3]. Remaining useful life (RUL) prediction, as an important component of PHM, can provide valuable insights for equipment maintenance, help organizations optimize maintenance strategies, reduce maintenance costs, and avoid economic losses caused by sudden equipment failures [4].

Rolling bearings are essential components in mechanical machinery, and their condition directly affects the entire performance and safety of the equipment. Therefore, precisely forecasting the RUL of bearings enhances equipment reliability and mitigates maintenance expenses to a certain degree [5, 6]. However, due to the intricate and nonlinear vibration signals produced during bearing operation, extracting effective degradation features and making precise predictions remain significant challenges in current research [7].

In recent years, numerous scholars have engaged in research on bearing RUL prediction methods, consistently exploring and developing new technologies and models. RUL prediction methods are classified into two main categories: model-based and data-driven approaches [8-10]. Model-based approaches estimate RUL by developing physical or mathematical models that effectively characterize the bearing degradation process, such as particle filtering [11], Weibull distribution [12], Kalman filtering [13], gamma processes [14] and Wiener processes [15]. The construction of such models requires not only a series of measurement parameters from actual engineering systems but also extensive prior knowledge. Although these methods are effective for predicting the overall trends of mechanical degradation, they often face challenges in practical industrial applications, particularly in accurately modeling the degradation processes of complex machinery using simple physical or mathematical models. The rapid advancement of artificial intelligence and deep learning has led to the emergence of data-driven approaches as the predominant method for predicting the RUL of bearings [16]. Consequently, the investigation of data-driven methods for predicting the RUL of bearings has become a prominent area of contemporary research aimed at achieving accurate RUL predictions [17].

The rapid advancement of intelligent sensing technologies and machine learning has resulted in the accumulation of substantial condition monitoring data in industrial production, facilitating the swift emergence of data-driven methods. Machine learning and deep learning are primary research trajectories in contemporary data-driven methodologies [18]. Guo et al. [19] introduced an innovative time-domain prediction framework utilizing empirical mode decomposition (EMD) and long short-term memory (LSTM) to enhance the precision and resilience of rolling bearing RUL prediction. Zhou et al. [20] introduced a rolling bearing RUL prediction framework based on EMD signal preprocessing and B-spline mother wavelet neural networks, which extracted degradation features through physics-guided signal preprocessing and established the correlation between degradation features and RUL using wavelet neural networks. However, this method relies on expert knowledge and experience during the feature extraction phase, and requires manual selection and development of suitable feature extraction techniques. Additionally, the feature extraction process did not fully consider spatial feature information. By contrast, deep learning models can automatically extract and learn more advanced abstract features from unstructured data, thereby removing the necessity for manual feature design and selection. Liu et al. [21] introduced a RUL prediction methodology predicated on dynamic operational conditions, which optimized vibration data to extract features with strong trend and noise resistance through adaptive variational mode decomposition, and combined one-dimensional convolutional LSTM with attention mechanism to achieve high-precision RUL prediction. Deng et al. [22] constructed a hybrid deep learning model based on continuous wavelet transform, multilayer perceptron, deep autoregressive models, and transformer for rolling bearing RUL prediction. Yang et al. [23] proposed a physics-driven multistate time-frequency bearing RUL prediction method. Although deep learning has shown outstanding accuracy and efficiency in RUL prediction, its application still faces several challenges, including substantial demand for computational resources and constrained performance in processing extensive time-series data. To resolve this matter, this paper introduces an efficient Mamba network [24], which optimizes the computational architecture and combines spatiotemporal deep feature extraction capabilities to reduce resource consumption while enhancing the processing performance of long time-series data, thereby achieving more accurate RUL predictions.

Most existing RUL prediction methods depend on data from a single sensor, which makes it difficult to effectively address the complexities introduced by multisensor fusion and non-stationary signals. Compared with single-sensor data, multisensor data can provide more comprehensive information about the equipment's operating conditions, offering greater research value for health management under complex operating conditions [25-27]. Yan et al. [28] extracted both time-domain and frequency-domain features from the horizontal and vertical directions of rolling bearings and constructed a comprehensive health indicator by fusing multichannel features to more accurately predict RUL. Zhang et al. [29] transformed multisensor monitoring data into grid and graph structures to predict RUL. Li et al. [30] built a hierarchical attention graph convolutional network with multisensor data, aiming to simultaneously model the geographical and temporal dependencies of the sensors to enhance the precision and efficacy of RUL prediction. However, many technical challenges remain in achieving efficient fusion and feature extraction from multisensor data under complex operating conditions and non-stationary signals. To overcome these limitations, this paper proposes a novel RUL prediction method for rolling bearings based on the Mamba-scaled dot product (SDP) framework. This method achieves efficient integration and feature extraction of multisensor data through modular design, thereby improving prediction accuracy and model robustness. The Mamba-SDP framework is specifically chosen in this work because it effectively addresses the primary challenges faced by existing RUL prediction methods, particularly in terms of long-range temporal dependency modeling, efficient multisensor information fusion, and robust feature extraction from non-stationary signals. Compared with conventional models, Mamba-SDP combines the strengths of advanced sequence modeling (Mamba module), channel-wise feature fusion, and attention-based normalization to ensure superior performance in both accuracy and generalization. This feature makes it especially suitable for real-world industrial scenarios characterized by high complexity, variable working conditions, and noise. By integrating these novel modules, the proposed method is able to capture both local and global degradation patterns, thus achieving more reliable and stable RUL prediction for rolling bearings. The primary contributions of this paper are as follows:

1) A multisensor data fusion method based on a channel fusion strategy is proposed to achieve efficient integration of multidimensional information. The Mamba module extracts spatiotemporal deep features from multisensor data to improve the assessment of equipment health condition.
2) The improved channel fast Fourier transform (ICFFT) module is developed to augment the model's capacity to process non-stationary signals through time-frequency

domain transformation. The designed SDP module, combined with cross-normalization (CNorm), reduces numerical fluctuations within the high-dimensional feature space, thereby enhancing the model's stability and computational efficiency when capturing both local and global dependencies.

3) The shallow features weighted and focused by the SDP module are integrated with the deep characteristics extracted by the Mamba and ICFFT modules through residual connections, enabling the effective fusion of multilevel features and improving the utilization of both prior and posterior feature information. The fused features are passed through fully connected layers to output the predicted RUL of the bearing. The model's prediction performance under complex operating conditions is validated using two datasets.

The remainder of this paper is structured as follows: Sec. 2 presents the related theories, Sec. 3 delineates the suggested methodology, Sec. 4 assesses the method's efficacy with two bearing datasets, and Sec. 5 concludes the paper.

2. Related theory

2.1 Self-attention

The encoder-decoder framework based on recurrent networks such as recurrent neural network (RNN) and LSTM cannot effectively utilize long-distance information by converting the input sequence into the final hidden state of the encoder, making it difficult to model long sequence data. The attention mechanism, however, can overcome this limitation by learning the weights of input information at different time steps. Fig. 1 illustrates a commonly used version of attention in time-series prediction, namely, self-attention.

First, given an input sequence $x=\{x_1,x_2,\cdots,x_n\},x_i\in R^k$, the query, key, and value vectors are obtained through linear transformations. With $q_i,k_i,v_i\in R^k$ taken as an example, the calculation process for each element is

$$q_i=W^q x_i,k_i=W^k x_i,v_i=W^v x_i,i=1,2,\cdots,n \tag{1}$$

where $W^q,W^k,W^v\in R^{d_k\times k}$ is a learnable matrix, and q_i,k_i,v_i represent the corresponding query, key, and value vectors, respectively. Then, with q_i and k_j used as examples, the similarity score is computed through the SDP function. The scores are then normalized using the softmax function, resulting in the weight $\hat{\alpha}_{i,j}$ for each sample point.

$$\alpha_{i,j}=\frac{q_i\bullet k_j}{\sqrt{d_k}} \tag{2}$$

where d_k is the scaled factor in the attention mechanism, which corresponds to the dimension of the key vector k_j. It is used to prevent gradient vanishing caused by excessive dot product values.

$$\hat{\alpha}_{i,j}=\frac{\exp(\alpha_{i,j})}{\sum_{t=1}^{n}\exp(\alpha_{i,t})} \tag{3}$$

Finally, the output z_i is computed as the weighted sum of the value vectors.

$$z_i=\sum_{j=1}^{n}\hat{\alpha}_{i,j}v_j \tag{4}$$

2.2 State space model

The state space model (SSM) is capable of modeling both univariate and multivariate time series of dynamic systems in the presence of non-stationarity, structural alterations, and irregular patterns. Its core is to capture system dynamics through a state-space framework and constrain parameters using structured matrices. As shown in Fig. 2, SSM consists of latent states and observation variables, which are coupled through state transition equations and observation equations. With the parameterization capability of structured matrices, SSM can efficiently perform training and inference, making it suitable for complex and dynamically changing scenarios.

A linear time-invariant system is formed by projecting the input

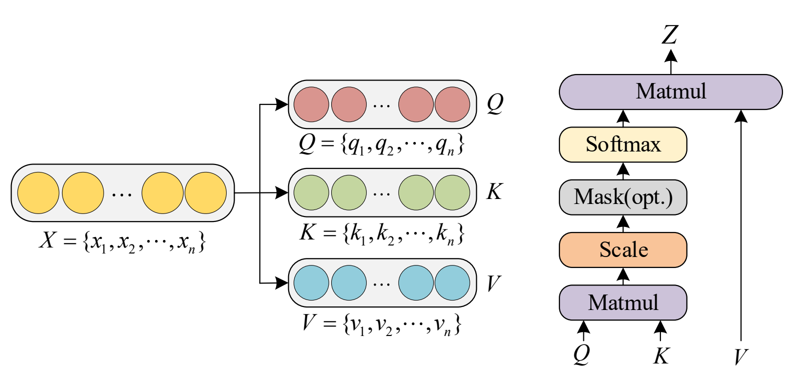


Fig. 1. Illustration of self-attention.

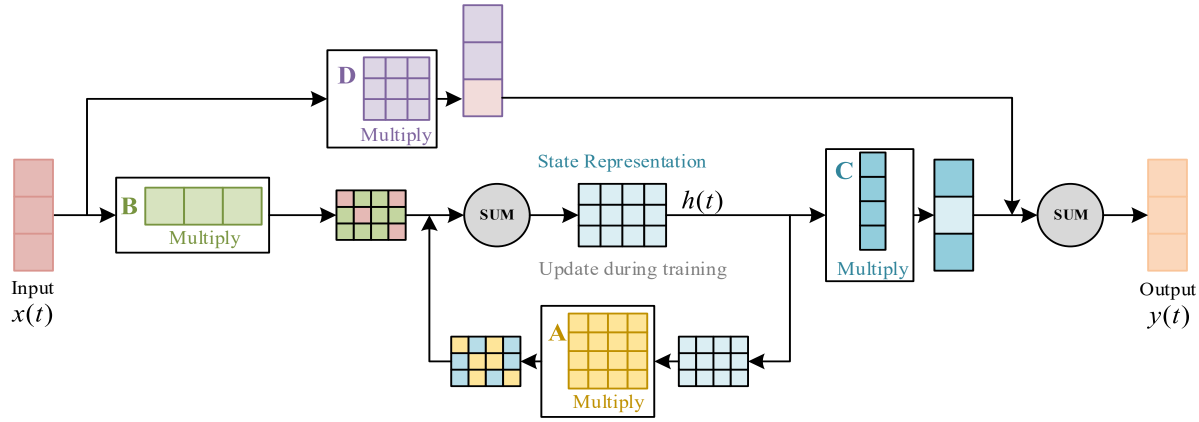


Fig. 2. Diagram of the SSM structure.

$x(t) \in \mathbb{R}^L$ and hidden state $h(t) \in \mathbb{R}^L$ onto the output $y(t) \in \mathbb{R}^L$. Mathematically, it can be expressed by a linear ordinary differential equation as follows:

$$\begin{aligned} h'(t) &= Ah(t) + Bx(t) \\ y(t) &= Ch(t) + Dx(t) \end{aligned} \tag{5}$$

where $A \in \mathbb{R}^{N\times N}$ represents the state transition matrix, which describes how the hidden state $h(t)$ updates over time and determines the impact of the present condition on state changes. $B \in \mathbb{R}^{N\times 1}$ represents the input matrix, which represents how the input $x(t)$ affects state changes and assesses the input's impact on state variation. $C \in \mathbb{R}^{1\times N}$ represents the output matrix, which depict how the hidden state maps to the output, determining the direct impact of the current state on the outcome. $D \in \mathbb{R}^{1\times 1}$ represents the direct transmission matrix, which describes how the input $x(t)$ directly affects the output $y(t)$, determining the direct contribution of the input to the output.

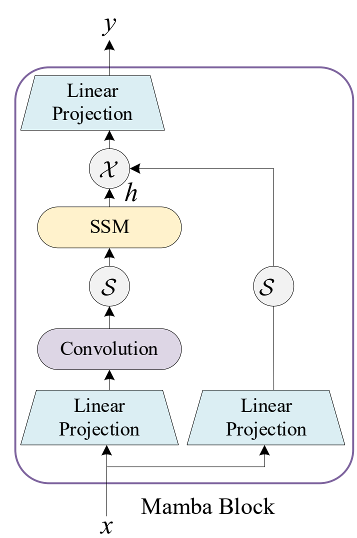


Fig. 3. Mamba architecture diagram.

2.3 Mamba

The Mamba (linear-time sequence modeling with selective state spaces) model represents an advanced and efficient sequence processing framework, designed on the basis of the principles of selective state space models. It is explicitly designed for processing long sequence data and can perform computations in linear time, demonstrating significant performance advantages over traditional transformers. The key improvement of Mamba lies in its dynamic parameter selection mechanism, which is dependent on the input, thereby allowing the model to selectively memorize or ignore specific information when processing long sequences. As a result, the problem of information propagation or forgetting in traditional state space models in long sequences is effectively solved. In addition, Mamba significantly reduces computational and memory requirements by reparameterizing the state matrix A and employing efficient parallel algorithms, making it run faster on hardware and applicable across various domains, such as cross-lingual tasks, audio processing, and genomics. The core highlight of Mamba is its state selection mechanism, which enables the model to adapt its internal state dynamically according to the input, selectively filtering or retaining important information. Fig. 3 illustrates the structure of Mamba.

The Mamba model combines the advantages of RNNs and convolutional neural networks (CNNs) to construct a structured state-space sequence model. Its state transition can be represented by the following equation:

$$h_{t+1} = Ah_t + Bx_t \tag{6}$$

$$y_t = Ch_t \tag{7}$$

where h_t denotes the concealed condition at a specific time step t; x_t is the input; and *A*, *B* and *C* are the model parameters. These parameters can dynamically adjust according to the input data to adapt to different features.

When using vibration data from rolling bearings for life prediction, the Mamba model can utilize its selective state space to effectively capture long-range dependencies in vibration signals, which is essential for accurately predicting the RUL of the bearing. By adjusting the state space of the Mamba model, it may more effectively capture the degrading feature information inside the vibration data, hence enhancing the accuracy and efficiency of the forecast.

3. Proposed method

3.1 Multisensor data fusion

Data processing is an essential phase in life span prediction. To minimize reliance on expert knowledge, this paper utilizes a multisensor information channel fusion strategy, which integrates vibration data from different locations and extracts multi-perspective features. As a result, it can comprehensively describe the bearing state, enrich the input feature information, and improve model performance. Fig. 4 illustrates the multisensor fusion process.

Assume that a dataset of degradation-to-failure data from *M* machines exists. Throughout the degrading process of each equipment, data are gathered from *C* channels of external sensors at a designated sample rate and documented at regular time intervals. Each degradation-to-failure data sequence is divided into multiple samples. Let $X_t^m \in \mathbb{R}^{H\times 2\times n}$ denote the t-th sample from the m-th machine, where H represents the sample length, 2 denotes the channel dimensions in the horizontal and vertical orientations, and n denotes the number of sensors. The dataset from the m-th machine can be represented as $\left\{\mathrm{x}_t^m\right\}_{t=1}^{n_m}$, where n_m denotes the quantity of samples obtained from the m-th machine.

Min-max normalization is initially employed to mitigate the effects of amplitude differences across several sensors during the processing of raw data. The equation is as follows:

$$\tilde{x}_{t,i}^c = \frac{x_{t,i}^c - \min(x_{t,i}^c)}{\max(x_{t,i}^c) - \min(x_{t,i}^c)} \tag{8}$$

where $x_{t,i}^c$ denotes the unprocessed data from the i-th sensor at the t-th temporal instance, and the normalized data are denoted as $\tilde{x}_{t,i}^c$. $\min(x_{t,i}^c)$ and $\max(x_{t,i}^c)$ represent the minimum and maximum values of the data, respectively. This method gives the data a uniform scale, which facilitates subsequent analysis. Next, the data from various channels are integrated to produce the multichannel fused data x_t^m:

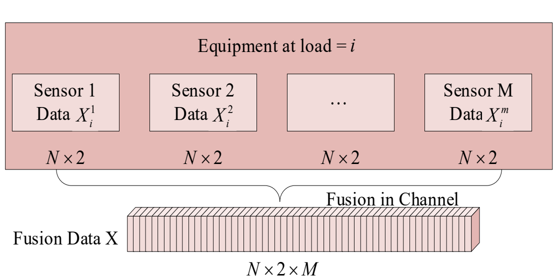


Fig. 4. Multisensor data fusion process based on channel fusion.

$$\mathrm{x}_t^m = \left\{\tilde{x}_t^1, \tilde{x}_t^2, \cdots, \tilde{x}_t^C\right\} \tag{9}$$

where $\mathrm{x}_t^m \in \mathbb{R}^{H\times 2\times m}$ represents the t-th fused sample from the m-th machine.

Let y_t^m represent the true RUL value of the t-th sample from the m-th machine, and $\left\{\mathrm{x}_t^m, y_t^m\right\}_{t=1}^{n_m}$ denote the training set of the m-th machine. These M training sets are utilized for supervised learning to train a model for predicting the RUL of new machines.

3.2 Scaled dot product attention mechanism module

The traditional self-attention mechanism, as shown in Fig. 1, generally assesses the significance of values by evaluating the similarity between the query *Q* and the key *K*. Nonetheless, when addressing lengthy sequences, the computational complexity of the traditional approach increases significantly. To optimize computational efficiency, this paper proposes an improved self-attention mechanism: the SDP attention module. In the SDP module, the results of *K* and *V* are precomputed and cached in advance under certain conditions of *Q* and *K*. When a new query *Q* arrives, *K* and *V* do not need to be recalculated; instead, a single matrix multiplication with the cached results is performed, thereby avoiding the repetitive computation of key-value pairs and reducing computational overhead and improving efficiency.

In addition, the SDP module adopts a CNorm operation, which normalizes Q_U, K_U, and V_U by using a linear kernel method before the matrix multiplication. This approach not only avoids the numerical instability caused by the softmax operation in the traditional attention mechanism but also mitigates the issue of numerical explosion when computing *Q*, *K*, and *V*. The architecture of the SDP attention mechanism module is illustrated in Fig. 5.

The equation for computing the SDP attention layer can be expressed as

$$U(x) = CN(Q_U)\cdot CN(K_U^T V_U) \tag{10}$$

$$\begin{aligned} Q_U &= \varphi(x) \\ K_U &= \gamma(x) \\ V_U &= \eta(x) \end{aligned} \tag{11}$$

where φ, γ, and η represent linear transformations and view operations. The calculation method for CNorm is shown in the following equation:

$$CN(x) = \frac{gx}{\sqrt{\sum_{i=0}^{h}\|x\|^2}} \tag{12}$$

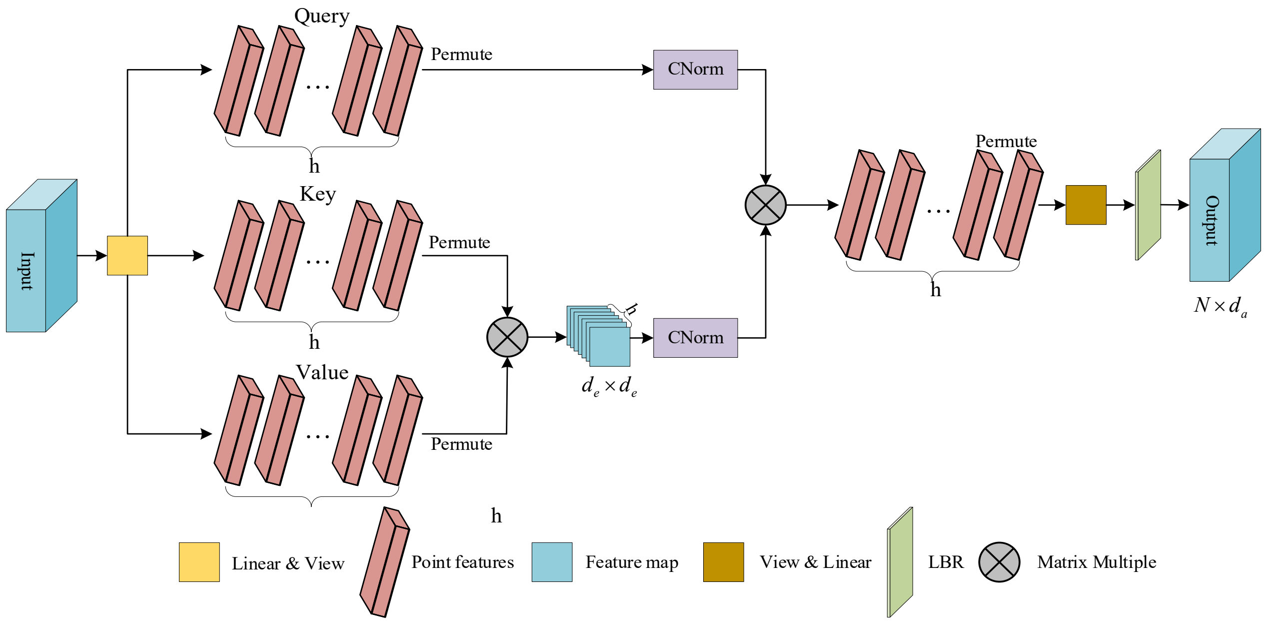


Fig. 5. Schematic representation of the SDP attention mechanism module.

where g is a learnable parameter with an initial value of a random matrix, and its embedding dimension is h. $CN(x)$ is generated using a linear kernel method to form h clusters and is normalized using the L2 norm. This norm is applied along two dimensions: the spatial dimension of K^TV and the channel dimension of Q, hence the term "cross-normalization."

$CN(x)$ is used to normalize K^TV, thereby obtaining $K^TV-Norm$. Meanwhile, $CN(x)$ is also used to normalize Q_U, generating Q_U_Norm. This method replaces the softmax in the traditional self-attention mechanism with CNorm, where the keys and values in the self-attention are directly multiplied.

$$[K^TV]_{i,j} = \sum_{k=1}^{n} K_{ik}^T V_{kj} \quad (13)$$

As shown in Fig. 5, CNorm is applied to both the query and the output.

$$U(x) = \begin{bmatrix} \hat{q}_o \cdot \hat{k}_0 & \hat{q}_o \cdot \hat{k}_1 & \cdots & \hat{q}_o \cdot \hat{k}_h \\ \hat{q}_1 \cdot \hat{k}_0 & \hat{q}_1 \cdot \hat{k}_1 & \cdots & \hat{q}_1 \cdot \hat{k}_h \\ \vdots & \vdots & \ddots & \vdots \\ \hat{q}_N \cdot \hat{k}_0 & \hat{q}_N \cdot \hat{k}_1 & \cdots & \hat{q}_N \cdot \hat{k}_h \end{bmatrix} \quad (14)$$

$$\hat{q}_i = CN[(Q_{i0}, Q_{i1}, \cdots, Q_{ih})] \quad (15)$$

$$\hat{k}_i = CN[([K^TV]_{0i}, [K^TV]_{1i}, \cdots, [K^TV]_{hi})] \quad (16)$$

In this attention mechanism, the projection weights are used to scale and aggregate the dot product terms through a weighted sum.

$$[W_{proj}U(x)]_{i,j} = \sum_{m=1}^{h} \omega_{mj} \hat{q}_i \cdot \hat{k}_j \quad (17)$$

In Eq. (17), the relational features are defined by the cosine similarity between different patches and clusters. CNorm constrains the features of each pixel in the query and clusters as unit vectors, and avoids suppressing relational attributes by normalizing them to a finite length. If these feature values are arbitrary, then the attention regions will depend on the initialization.

3.3 Improved channel fast Fourier transform

Fast Fourier transform (FFT) [31] is an efficient algorithm for the discrete Fourier transform (DFT) that turns time-domain information into frequency-domain signals, elucidating the frequency components and energy distribution of the signal. It is an essential tool in signal processing. However, FFT has several limitations. First, it assumes that the signal is stationary within the analysis window, complicating the management of non-stationary signals. Second, the transformation to the frequency domain results in the loss of time-domain information, which prevents time-frequency joint analysis. Third, FFT is sensitive to noise and is prone to interference, leading to distortions in the frequency spectrum.

To overcome these limitations, this paper proposes an improved channel frequency-domain modeling method, termed ICFFT. ICFFT uses channel matrix multiplication (CMM) to independently process real and imaginary components in the frequency domain and then recombine them into a complex form, thereby achieving precise modeling and extraction of frequency-domain information. ICFFT is particularly suitable for analyzing non-stationary periodic features in rolling bearing vibration signals. By utilizing both frequency-domain and time-domain transformations, it can improve the capacity to identify degradation characteristics and the precision of RUL forecasts. Specifically, the ICFFT has three primary steps:

(1) Spectral transformation: First, the time-domain signal is transformed into a frequency-domain signal via the DFT. Through the DFT, the signal is partitioned into real and imaginary components, each corresponding to distinct frequency elements within the signal. The equation for DFT is given as

$$F(k)=\int_D f(x)e^{-j2\pi kx}dx=\int_D f(x)\cos(2\pi kx)dx + j\int_D f(x)\sin(2\pi kx)dx \tag{18}$$

where k is the frequency variable, x is the spatial variable, j is the imaginary unit, $\int_D f(x)\cos(2\pi kx)dx$ is the real part, and $j\int_D f(x)\sin(2\pi kx)dx$ is the imaginary part.

(2) CMM: In the frequency domain, the real and imaginary parts are mixed through CMM. This procedure augments the extraction of frequency-domain information and subsequently enhances the model's capacity to differentiate and recognize various frequency components. In this way, ICFFT can accurately process complex non-stationary signals and strengthen the key frequency features within the signal.

(3) Inverse spectral transformation: After the frequency-domain feature extraction is completed, inverse FFT (IFFT) is employed to revert the frequency-domain signal back into the time-domain signal. The equation for the IFFT is given as

$$f(x)=\int_D F(k)e^{j2\pi kx}df \tag{19}$$

where k is the frequency variable, and $e^{j2\pi kx}$ is the kernel function of the Fourier transform, which is used to convert the frequency-domain signal back into the time-domain. This stage transforms the features obtained in the frequency domain back into a time-domain signal, completing the overall signal processing. Through this conversion between the frequency and time domains, ICFFT can effectively capture the periodic and global features in the time series.

3.4 Construction of the Mamba-SDP lifetime prediction model

This paper presents a Mamba-SDP-based model for predicting the RUL of rolling bearings, comprising three primary stages: data collecting and preprocessing, feature extraction, and RUL prediction with result visualization, as illustrated in Fig. 6.

First, in the data collection and preprocessing stage, data from the entire life cycle of the bearing are collected using multi-channel sensors, and channel fusion is performed to ensure the integrity and consistency of features across different sensors, which forms a solid foundation for subsequent feature learning. The fused data are then processed using min-max normalization to reduce noise influence and achieve standardization, thereby improving the model's robustness and the quality of data input.

Next, in the feature learning stage, the input data undergo LayerNorm normalization preprocessing, followed by the use of the Mamba module to extract spatiotemporal deep features. This module integrates the advantages of RNNs and CNNs, effectively capturing the temporal dependencies and spatial structures within the data. Meanwhile, dropout technology is applied to mitigate overfitting issues. By introducing residual connections, the model ensures that input information is effectively preserved during the feature extraction process. Subsequently, the ICFFT module combines dual transformations between time and frequency domains. It converts the time-domain signal into the frequency domain to extract key frequency features and captures periodic and local characteristics of degradation signals through spectral decomposition. At the same time, the IFFT is utilized to convert frequency-domain characteristics back to the time domain, thus achieving feature reconstruction and deep fusion of time-frequency domain features. Additionally, the designed residual connection further boosts the model's feature processing capability and robustness.

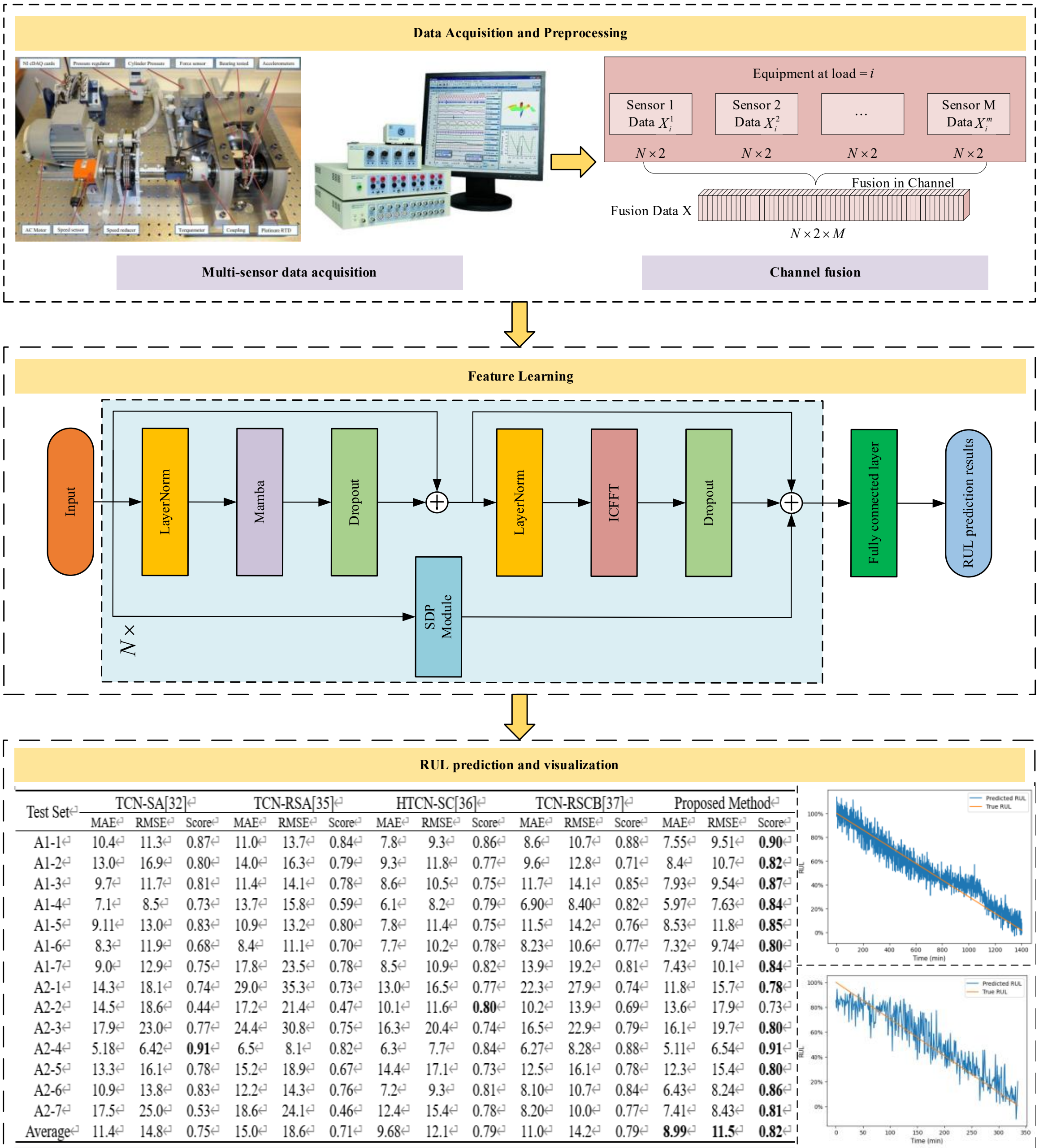
The SDP module, in conjunction with CNorm, mitigates numerical fluctuations in the high-dimensional feature space, thereby improving the model's stability and computational efficiency of the model when modeling local and global dependencies. To achieve effective fusion of multi-level features, the model combines shallow features, weighted and focused by the SDP module, with deeper features extracted by the Mamba and ICFFT modules using residual connections, improving the feature representation capability and the utilization efficiency of preceding and succeeding information. Ultimately, the fused features are mapped to RUL prediction values via a fully connected layer.

Finally, during the RUL prediction and visualization phase, the model performance is comprehensively evaluated using evaluation metrics such as mean absolute error (MAE), root mean square error (RMSE), and score. These metrics quantify the error between the predicted results and the true values while visualizing the curves to show the fitting performance between the real degradation trend and the predicted results. The prediction accuracy, robustness, and stability of the proposed method are verified from multiple perspectives, providing reliable evidence for its practical application.

3.5 RUL prediction

This study uses full life cycle data of bearings for training for the following reasons: (1) Full life cycle data cover all potential faults, aiding in comprehensive monitoring; (2) The interval between first damage and failure in bearings is brief, and early RUL prediction allows adequate time for maintenance; (3) It avoids the difficulty of manually determining the first prediction time.

Three metrics are used to assess model prediction perform-



Test Set	TCN-SA[32]			TCN-RSA[35]			HTCN-SC[36]			TCN-RSCB[37]			Proposed Method		
	MAE	RMSE	Score	MAE	RMSE	Score	MAE	RMSE	Score	MAE	RMSE	Score	MAE	RMSE	Score
A1-1	10.4	11.3	0.87	11.0	13.7	0.84	7.8	9.3	0.86	8.6	10.7	0.88	7.55	9.51	**0.90**
A1-2	13.0	16.9	0.80	14.0	16.3	0.79	9.3	11.8	0.77	9.6	12.8	0.71	8.4	10.7	**0.82**
A1-3	9.7	11.7	0.81	11.4	14.1	0.78	8.6	10.5	0.75	11.7	14.1	0.85	7.93	9.54	**0.87**
A1-4	7.1	8.5	0.73	13.7	15.8	0.59	6.1	8.2	0.79	6.90	8.40	0.82	5.97	7.63	**0.84**
A1-5	9.11	13.0	0.83	10.9	13.2	0.80	7.8	11.4	0.75	11.5	14.2	0.76	8.53	11.8	**0.85**
A1-6	8.3	11.9	0.68	8.4	11.1	0.70	7.7	10.2	0.78	8.23	10.6	0.77	7.32	9.74	**0.80**
A1-7	9.0	12.9	0.75	17.8	23.5	0.78	8.5	10.9	0.82	13.9	19.2	0.81	7.43	10.1	**0.84**
A2-1	14.3	18.1	0.74	29.0	35.3	0.73	13.0	16.5	0.77	22.3	27.9	0.74	11.8	15.7	**0.78**
A2-2	14.5	18.6	0.44	17.2	21.4	0.47	10.1	11.6	**0.80**	10.2	13.9	0.69	13.6	17.9	0.73
A2-3	17.9	23.0	0.77	24.4	30.8	0.75	16.3	20.4	0.74	16.5	22.9	0.79	16.1	19.7	**0.80**
A2-4	5.18	6.42	**0.91**	6.5	8.1	0.82	6.3	7.7	0.84	6.27	8.28	0.88	5.11	6.54	**0.91**
A2-5	13.3	16.1	0.78	15.2	18.9	0.67	14.4	17.1	0.73	12.5	16.1	0.78	12.3	15.4	**0.80**
A2-6	10.9	13.8	0.83	12.2	14.3	0.76	7.2	9.3	0.81	8.10	10.7	0.84	6.43	8.24	**0.86**
A2-7	17.5	25.0	0.53	18.6	24.1	0.46	12.4	15.4	0.78	8.20	10.0	0.77	7.41	8.43	**0.81**
Average	11.4	14.8	0.75	15.0	18.6	0.71	9.68	12.1	0.79	11.0	14.2	0.79	**8.99**	**11.5**	**0.82**

Fig. 6. Mamba-SDP lifetime prediction model.

ance: MAE, RMSE, and the designed score. The calculations for MAE and RMSE are given by Eqs. (21) and (22), respectively, where RUL_t represents the true RUL value of the rolling bearing at time t; $\overline{RUL_t}$ represents the predicted RUL value of the rolling bearing at time t; and e is the number of samples in the test set.

$$E_t = RUL_t - \overline{RUL_t} \tag{20}$$

$$MAE = \frac{1}{e}\sum_{i=1}^{n}|E_t| \tag{21}$$

$$RMSE = \sqrt{\frac{1}{e}\sum_{i=1}^{n}(E_t)^2} \tag{22}$$

In practical working scenarios, underestimation and overestimation of RUL predictions can have varying degrees of impact on the operation of mechanical equipment. Underestimation may lead to unnecessary downtime, while overestimation could result in equipment failure and even endanger production operators' safety. The scoring function needs to take into account the effects of underestimation and overestimation, as well as the characteristics of different stages in the equipment life cycle to comprehensively and objectively evaluate the prediction model. Therefore, in this study, the prediction weight is reduced in the early stages of the machine life cycle and increased in the later stages to more accurately capture faults and failures.

The scoring function used in this paper is defined on the basis of the work in Ref. [32]. This scoring function thoroughly evaluates the influence of different operational stages in the early and late phases of the equipment life cycle, as well as throughout the entire life cycle. The specific definitions are given by Eqs. (23) and (24)

$$A_t = \begin{cases} \exp(-\ln(0.6)\cdot(\frac{E_t}{10})), E_t \leqslant 0 \\ \exp(\ln(0.6)\cdot(\frac{E_t}{40})), E_t > 0 \end{cases} \tag{23}$$

$$\text{Score} = \alpha\frac{1}{m}\sum_{t=1}^{m} A_t + \beta\frac{1}{n-m}\sum_{m+1}^{n} A_t \tag{24}$$

where α and β are the early and late bearing life span weights, m is the early-stage percentage, A_t denotes the weighted error between the predicted and actual RUL values at time step t, $E_t \leq 0$ represents overestimation (i.e., RUL prediction in the later stages), and $E_t > 0$ represents underestimation (i.e., RUL prediction in the early stages). In this study, α is set to 0.35, and β is set to 0.65, indicating that predictions in the later stages of the life cycle are more important than those in the earlier stages. The score ranges from 0 to 1, with elevated values signifying superior model performance.

4. Experimental validation and analysis

To validate the effectiveness of the Mamba-SDP model, this paper uses the IEEE PHM2012 [33] and XJTU-SY Bearing Dataset [34] for evaluation. All experiments were conducted using the deep learning framework Torch, with the hardware environment consisting of an i7-14700KF processor, NVIDIA GeForce RTX 4070Ti Super, and 32 GB RAM.

4.1 Case study 1: Bearing RUL prediction using the PHM 2012 dataset

4.1.1 Introduction to the experimental datasets

The PHM 2012 dataset, published by the FEMTO-ST Institute, comprises bearing data ranging from the initial normal condition to failure. The data were gathered via the PRONOSTIA platform, as illustrated in Fig. 7. The data were sampled at 25.6 kHz, with recordings every 10 seconds, each lasting 0.1 seconds. The dataset used includes two operating conditions, which are described in Table 1.

4.1.2 Parameter configuration of the Mamba-SDP network

In the Mamba-SDP network, the corresponding hyperparameters need to be configured, as shown in Table 2. These parameters are established by cross-validation across various training datasets, with consideration of the prediction accuracy.

4.1.3 Experimental prediction results and discussion

This paper analyzes bearing vibration data under operating conditions 1 and 2 to thoroughly assess the performance of the Mamba-SDP network. Given both horizontal and vertical vibration signals provide extensive deterioration information, a multi-sensor data fusion approach is utilized to derive more complete, precise, and dependable features. Meanwhile, the Mamba-SDP feature learning network is used to deeply mine degradation features from both time-domain and frequency-domain perspectives. For each operating condition, one bearing dataset is chosen for testing, while the remaining datasets are used for network training. Furthermore, the RUL prediction results of the Mamba-SDP model are compared with those of four other methods, as shown in Table 3.

This experiment comprehensively evaluates the performance of 14 test bearings, and it evaluates techniques on three important metrics: MAE, RMSE, and score, as shown in Table 3. The MAE and RMSE metrics reflect the average and extreme

Table 1. Operating conditions for the FEMTO-ST bearing dataset.

Operating conditions	Rotational speed (r/min)	Load force (N)	Test bearing
1	1800	4000	A1-1–A1-7
2	1650	4200	A2-1–A2-7

Table 2. Mamba-SDP network structure parameter details.

Layer	Mamba block	ICFFT	SDP
Parameter settings	d_model:2 d_state:64 expand:2 d_conv:4 conv_bias: True	hidden_size:768 num_blocks:2 sparsity_threshold:0.01 scale: 0.02	d_model:2 d_k:2 d_v:2 h:8 dropout:0.1 gamma: Learnable (1,8,1,1)

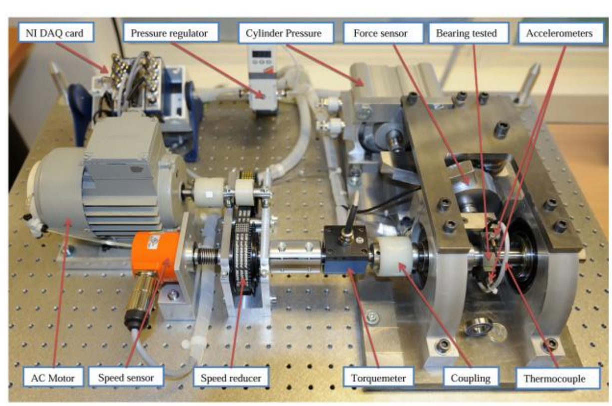


Fig. 7. Platform of PRONOTIA.

Table 3. Prediction results of different methods based on PHM 2012.

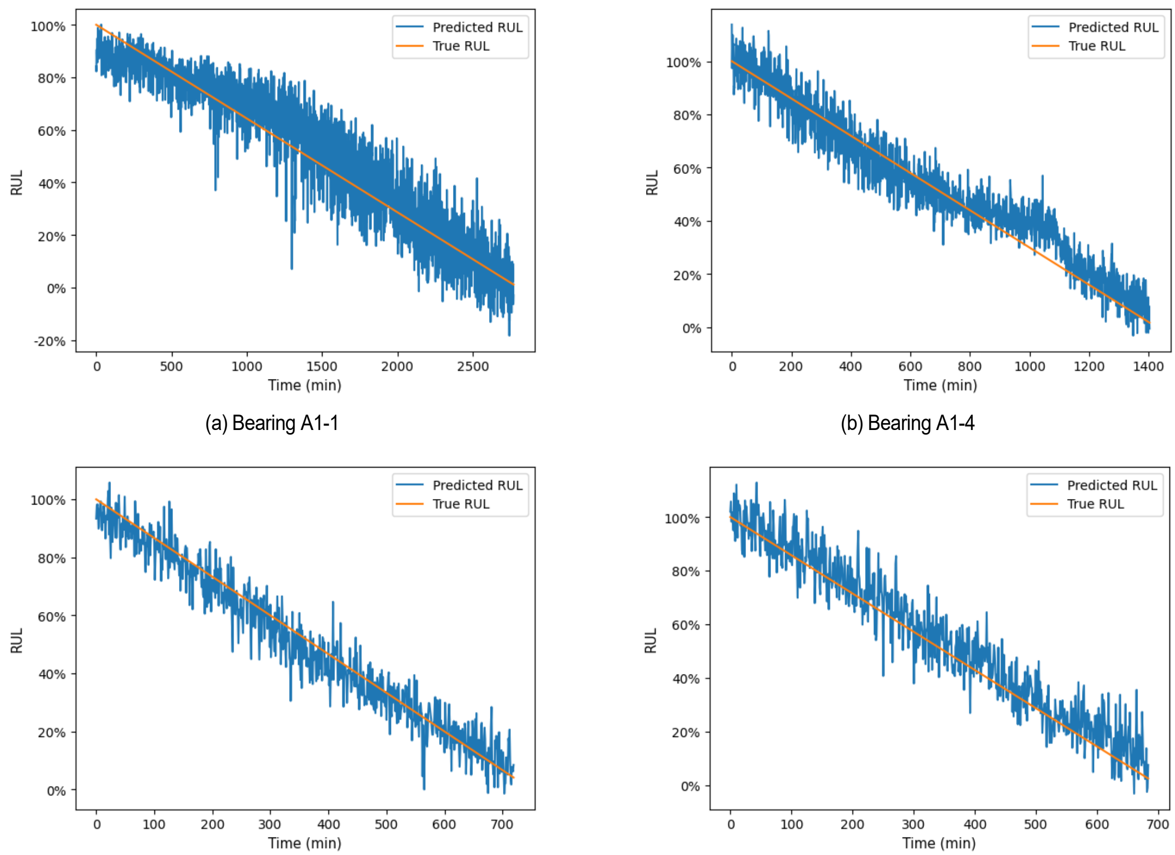
Test set	TCN-SA [32]			TCN-RSA [35]			HTCN-SC [36]			TCN-RSCB [37]			Proposed method		
	MAE	RMSE	Score	MAE	RMSE	Score	MAE	RMSE	Score	MAE	RMSE	Score	MAE	RMSE	Score
A1-1	10.4	11.3	0.87	11.0	13.7	0.84	7.8	9.3	0.86	8.6	10.7	0.88	7.55	9.51	**0.90**
A1-2	13.0	16.9	0.80	14.0	16.3	0.79	9.3	11.8	0.77	9.6	12.8	0.71	8.4	10.7	**0.82**
A1-3	9.7	11.7	0.81	11.4	14.1	0.78	8.6	10.5	0.75	11.7	14.1	0.85	7.93	9.54	**0.87**
A1-4	7.1	8.5	0.73	13.7	15.8	0.59	6.1	8.2	0.79	6.90	8.40	0.82	5.97	7.63	**0.84**
A1-5	9.11	13.0	0.83	10.9	13.2	0.80	7.8	11.4	0.75	11.5	14.2	0.76	8.53	11.8	**0.85**
A1-6	8.3	11.9	0.68	8.4	11.1	0.70	7.7	10.2	0.78	8.23	10.6	0.77	7.32	9.74	**0.80**
A1-7	9.0	12.9	0.75	17.8	23.5	0.78	8.5	10.9	0.82	13.9	19.2	0.81	7.43	10.1	**0.84**
A2-1	14.3	18.1	0.74	29.0	35.3	0.73	13.0	16.5	0.77	22.3	27.9	0.74	11.8	15.7	**0.78**
A2-2	14.5	18.6	0.44	17.2	21.4	0.47	10.1	11.6	**0.80**	10.2	13.9	0.69	13.6	17.9	0.73
A2-3	17.9	23.0	0.77	24.4	30.8	0.75	16.3	20.4	0.74	16.5	22.9	0.79	16.1	19.7	**0.80**
A2-4	5.18	6.42	**0.91**	6.5	8.1	0.82	6.3	7.7	0.84	6.27	8.28	0.88	5.11	6.54	**0.91**
A2-5	13.3	16.1	0.78	15.2	18.9	0.67	14.4	17.1	0.73	12.5	16.1	0.78	12.3	15.4	**0.80**
A2-6	10.9	13.8	0.83	12.2	14.3	0.76	7.2	9.3	0.81	8.10	10.7	0.84	6.43	8.24	**0.86**
A2-7	17.5	25.0	0.53	18.6	24.1	0.46	12.4	15.4	0.78	8.20	10.0	0.77	7.41	8.43	**0.81**
Average	11.4	14.8	0.75	15.0	18.6	0.71	9.68	12.1	0.79	11.0	14.2	0.79	8.99	11.5	**0.82**

prediction errors, respectively, directly measuring the accuracy and reliability of the model.

The score metric comprehensively evaluates both the accuracy and timeliness of RUL prediction and is particularly sensitive to prediction quality in later degradation stages, making it a crucial indicator for practical application. Experimental results show that Mamba-SDP predicts well on most test datasets. Its average MAE is 8.99, significantly lower than that of TCN-RSA (15.0); the average RMSE is 11.5, which is notably lower than that of TCN-SA (14.8). This finding indicates that Mamba-SDP offers a significant advantage in reducing both mean and maximum prediction errors, effectively improving the overall prediction accuracy and consistency. Moreover, the average score of Mamba-SDP is 0.82, which is the highest among all the compared methods, reflecting its strong generalization capability and high stability in complex dynamic scenarios. The consistently high score suggests that the method is able to maintain excellent predictive performance not only in average cases but also when timely and accurate end-of-life predictions are required. In specific tests, the Mamba-SDP method outperforms other methods in different test sets. For example, in the A1-4 test set, its RMSE is 7.63, significantly better than that of TCN-SA (8.5) and of TCN-RSA (15.8). It has a score of 0.84, which is higher than that of HTCN-SC (0.79) and of TCN-RSCB (0.82). In the A2-4 test set, its RMSE further decreases to 6.54, outperforming TCN-SA (6.42) and HTCN-SC (7.7), while its score reaches 0.91, the highest among all methods. Even in the more challenging A2-5 test set, Mamba-SDP maintains the lowest RMSE and the highest score, further demonstrating the method's robustness in difficult scenarios. Although some compared methods perform better on specific test sets, such as TCN-SA, which achieves a score of 0.91 in the A2-4 dataset, and HTCN-SC, which achieves a score of 0.80 in the A2-2 dataset, they tend to exhibit greater error fluctuations and instability across the broader set of test cases. By contrast, Mamba-SDP demonstrates outstanding overall performance and stronger robustness due to its continuous leading advantage in multiple key performance metrics. Moreover, in some extreme cases, the performance gap between the Mamba-SDP model and comparison methods becomes even more pronounced, especially in scenarios with complex degradation patterns or high noise. These results confirm that the Mamba-SDP method can reliably capture essential degradation features throughout the entire bearing life cycle and maintain stable prediction accuracy under various operating conditions. Overall, the Mamba-SDP method not only leads in key indicators such as MAE, RMSE, and score but also demonstrates strong robustness and adaptability when dealing with complex data scenarios. Its advantages are particularly prominent in capturing degradation features throughout the entire bearing life cycle, offering a significant competitive edge for RUL prediction tasks. These characteristics further confirm that the Mamba-SDP method has significant progressiveness and potential value both in theory and practical applications, providing reliable support for fault prediction and health management in complex industrial environments.

Fig. 8 illustrates RUL predictions for bearings A1-1, A1-4, A2-4, and A2-6. The results comprehensively reflect the predictive performance and adaptability of the proposed Mamba-SDP model in various testing scenarios. Overall, the method accurately fits the actual degradation trend of the equipment, demonstrating high prediction accuracy, robustness, and broad applicability.

Notably, as shown in Fig. 8, while the predicted RUL curves generally follow the true degradation trend, some fluctuations and noise are present in the predicted values, especially during the early and middle stages of the bearing life cycle. These fluctuations can be attributed to several factors: (1) the inherent



(a) Bearing A1-1 (b) Bearing A1-4

(c) Bearing A2-4 (d) Bearing A2-6

Fig. 8. Visualization of RUL prediction results for test bearings in the PHM 2012.

noise in the raw vibration signals collected under real-world operating conditions, such as sensor interference and environmental disturbances; (2) the weak and less distinguishable degradation features in the early stages, which make the model more sensitive and prone to prediction jitter at this phase; (3) the necessity to balance the model's stability and generalization across diverse and sometimes limited training scenarios, which can also introduce minor deviations; and (4) the possible signal drift caused by dynamic changes in external conditions, such as load and temperature, which further increases the prediction difficulty.

In the A1-1 test set, the predicted values generally follow the actual RUL trend. Although noticeable fluctuations occur in the early and middle stages—mainly due to weak degradation features and increased sensitivity to noise at these phases—the prediction error gradually decreases as degradation progresses, indicating the model's ability to extract clearer features in the later stages. For the A1-4 test set, the predicted RUL values almost completely match the ground truth throughout the degradation process, especially in the later stages, reflecting both high fitting accuracy and strong model stability under relatively ideal signal conditions. In A2-4, the model maintains a high degree of consistency between predicted and true values across the full life cycle, demonstrating good generalization to complex degradation patterns. This finding indicates that the model can effectively handle varying and nonlinear characteristics in real-world data. In A2-6, although the prediction curve shows more fluctuations and a slight decrease in fitting accuracy at the late stage —possibly due to stronger noise, signal drift, or operating condition changes—the overall trend remains aligned with the true RUL. The prediction errors are still controlled within a reasonable range, demonstrating that the proposed model possesses considerable robustness even under challenging and dynamic operating environments.

In summary, the Mamba-SDP model exhibits stable and accurate prediction performance across multiple test scenarios. The results confirm its ability to adapt to both simple and complex degradation processes, as well as to maintain reliable generalization and robustness under various real-world conditions.

4.1.4 Ablation experiment

Ablation tests were performed by eliminating or altering one portion of the suggested approach to test its novel components.

The experiment used the A2-4 test bearing. The proposed technique used multisensor fusion data, while method 1 used vertical vibration data from a single vertical sensor. Method 2 extracted network features directly using the Mamba block, while the suggested technique used SDP-Mamba-ICFFT. Method 3 used the self-attention mechanism for feature fusion, while the

Table 4. Ablation experiment results of different comparison methods.

	Method 1	Method 2	Method 3	Proposed method
MAE	15.7	11.89	8.83	5.11
RMSE	20.2	14.57	10.8	6.54

suggested technique used the SDP attention mechanism module for adaptive weighted output feature fusion. This section compares these four methods through MAE and RMSE indicators, with results shown in Table 4.

Table 4 shows that the suggested model outperforms alternative techniques. Compared with existing approaches, the proposed method reduced MAE by 67 %, 57 %, and 42 %, and RMSE similarly. This improvement is attributed to the fusion of multisensor data, which provides more comprehensive signal features, thereby enhancing the model's ability to express complex signals and capture periodic characteristics. The SDP-Mamba-ICFFT model significantly improves prediction accuracy by independently modeling the real and imaginary parts in the frequency domain and optimizing the feature extraction process using the attention mechanism. Additionally, the SDP module, combined with CNorm, effectively reduces numerical fluctuations in the high-dimensional feature space, enhancing the model's stability and computational efficiency when capturing local and global dependencies. The final results show that the suggested strategy considerably improves input feature modeling, feature extraction, and feature fusion model performance.

4.2 Case study 2: Bearing RUL prediction using the XJTU-SY dataset

Fig. 9 shows the test equipment for the XJTU-SY Rolling Bearing Accelerated Life Test Dataset. Radial force is applied by a hydraulic loading system, and rotational speed is controlled by an AC motor speed controller. The vibration signals are horizontal and vertical, sampled at 25.6 kHz, 1 minute apart, and 1.28 seconds each. The dataset includes two operating conditions, with detailed information provided in Table 5.

4.2.1 Analysis of prediction results

For a comprehensive assessment of the prediction network's generalization capability, bearing vibration data from operating condition 1 and 2 were utilized in the experiment. The network parameters—MAE, RMSE, and score function—for the Mamba-SDP model were consistent with those utilized in case 1 when employing the XJTU-SY dataset for RUL prediction. The generalization capability of the proposed method was illustrated through a comparison between the prediction results of the Mamba-SDP model and four other methods from case 1: TCN-SA, TCN-RSA, HTCN-SC, and TCN-RSCB. Fig. 10 displays the results of the prediction results.

Fig. 10 clearly highlights the superior prediction performance of the proposed method. In terms of MAE and RMSE, the proposed method shows significantly lower error values, indicating its remarkable ability to accurately capture the bearing degradation trend and effectively control prediction errors. The score metric is defined as follows: The Mamba-SDP achieves high scores in all experiments, further validating its comprehensive prediction performance and stability under complex operating conditions.

Table 5. Operating conditions for the SJTU-SY bearing dataset.

Operating conditions	Rotational speed (r/min)	Load force (kN)	Test bearing
1	2100	12	B1-1–B1-5
2	2250	11	B2-1–B2-5

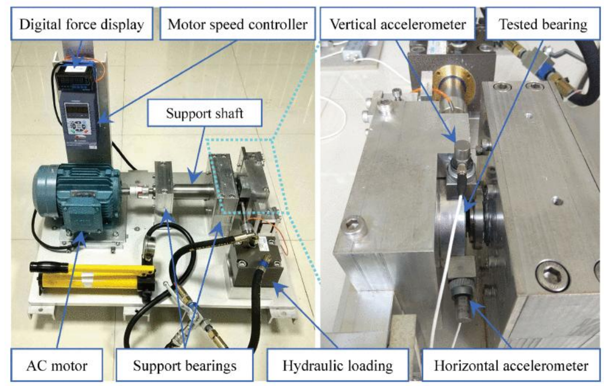
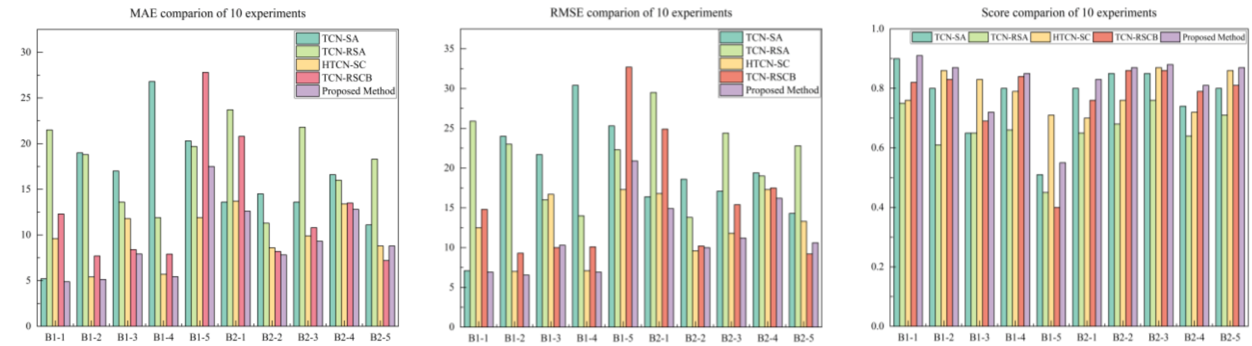


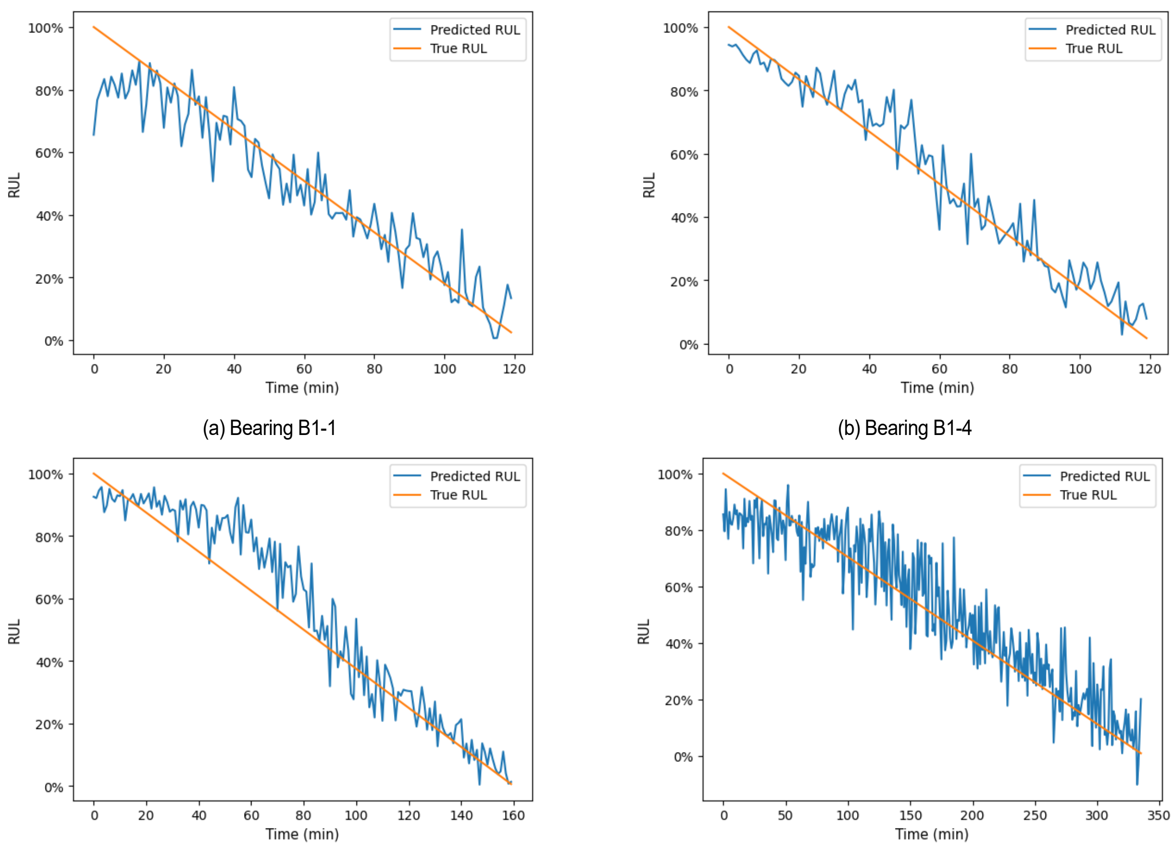
Fig. 9. Bearing test rig of XJTU-SY bearing dataset.

Fig. 11 displays the RUL prediction results for bearings B1-1, B1-4, B2-2, and B2-5, further verifying the prediction capability and generalization of the Mamba-SDP method under various operating conditions. Although some local fluctuations occur in the early degradation stage due to noise and the complexity of the operating conditions, the model overall accurately fits the actual RUL, especially in the later life stages where the predicted results closely match the true values. For instance, the prediction curves of B1-4 and B2-2 demonstrate that the model smoothly tracks the bearing's degradation trend throughout its entire life cycle, showing exceptional accuracy in late-stage prediction and global trend capture ability. Notably, however, the prediction error is relatively higher for B2-2 (Fig. 11(c)) than for other cases, especially in the early and middle stages. This higher error can be attributed to the more complex and abrupt degradation patterns under the specific operating environment of B2-2, as well as higher variability in load, speed, or external disturbances, which leads to non-stationary and noisier vibration signals. These factors increase the modeling challenge and may result in transient instability in the predicted RUL curve. Nevertheless, the Mamba-SDP model is still able to accurately track the overall degradation trend in the later stage, reflecting its robustness and adaptability under challenging conditions. For B1-1 and B2-5, although the early predictions show larger fluctuations, the model gradually converges and approaches the real RUL in the later



(a) MAE comparison (b) RMSE comparison (c) Score comparison

Fig. 10. MAE, RMSE, and score comparisons of SJTU-SY bearing datasets.



(a) Bearing B1-1 (b) Bearing B1-4

(c) Bearing B2-2 (d) Bearing B2-5

Fig. 11. Visualization of RUL prediction results for test bearings in the XJTU-SY.

stages, further proving its ability to capture complex degradation signals globally.

Figs. 10 and 11 demonstrate that the Mamba-SDP method outperforms other methods in various experimental settings, showing strong adaptability and the ability to capture degradation characteristics across the entire life cycle. The method exhibits high reliability in the later life stages, validating its technical advancements, robustness, and generalization ability. Therefore, it is highly applicable for industrial equipment life prediction and predictive maintenance.

4.2.2 Ablation experiment

The generalization capability of the proposed method across various datasets was assessed through an ablation experiment that was conducted similarly to the approach taken with the PHM 2012 dataset, utilizing bearing B1-4 for testing purposes. The prediction results are shown in Table 6. According to Table 6, the proposed method outperforms the compared methods in terms

Table 6. Ablation experiment results of different compared methods.

	Method 1	Method 2	Method 3	Proposed method
MAE	17.9	11.7	8.47	5.43
RMSE	22.6	13.6	9.21	6.91

of both MAE and RMSE, particularly demonstrating stronger robustness when handling variations in signal operating conditions. This finding indicates that the proposed multisensor fusion and RUL prediction method exhibits good adaptability under different testing conditions. The designed SDP module combined with CNorm further stabilizes the information representation in the high-dimensional feature space, significantly enhancing the model's ability to capture both global dependencies and local details. The experimental results demonstrate the reliability and applicability of the method across new tasks and scenarios.

5. Conclusions

To overcome the limitations of conventional models in global feature extraction and multisensor data fusion, this study proposes a novel rolling bearing RUL prediction model based on the Mamba-SDP framework. The Mamba-SDP framework is selected for its advanced ability to capture long-range temporal dependencies, efficiently fuse information from multiple sensors, and extract robust features from both stationary and non-stationary signals. By leveraging synergistic multimodule collaboration, the proposed model effectively integrates multidimensional information across frequency, time, and sensor domains, substantially improving the accuracy, generalization, and reliability of RUL prediction in complex and variable operational environments. The main conclusions of this work are as follows:

1) A data channel fusion method is presented. This method fuses multisensor data via channels to efficiently integrate it and overcome the constraints of single-sensor prediction approaches.
2) A modular design is adopted, integrating various components such as the Mamba module, the ICFFT module, and the SDP module to work collaboratively. This design allows the model to efficiently capture both local and global dependencies, enhancing prediction accuracy for long time series data. Under complex operating conditions, the model may more thoroughly encapsulate the deterioration characteristics of bearings. Meanwhile, the SDP module suppresses numerical fluctuations in the high-dimensional feature space, further improving the stability and robustness of the model.
3) The efficacy and generalization capacity of the Mamba-SDP model were corroborated using two datasets. Experimental results demonstrate that the Mamba-SDP model exhibits outstanding robustness in bearing degradation prediction under diverse and challenging operating conditions. Compared with other models, Mamba-SDP consistently achieves superior accuracy, stability, and strong applicability, particularly excelling in long-term sequence prediction scenarios.

The Mamba-SDP method is adopted in this study because it can address the key shortcomings of conventional and existing deep learning models, especially in terms of global spatiotemporal feature extraction and multisensor fusion. This approach provides a significant improvement for practical industrial health management and maintenance decision-making.

Although the Mamba-SDP model performs well in bearing degradation prediction in this study, it still has some limitations. As only vibration signals are used, vibration data have certain constraints in the representation of the degradation process under multiple operating conditions. Future research could incorporate additional sensor data, such as temperature and acoustic signals, to more comprehensively describe the degradation process of bearings, thereby further improving the model's prediction accuracy and robustness under various operating conditions.

Acknowledgments

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Conflict of interest

The authors declare no competing interests.

Nomenclature

RUL : Remaining useful life
PHM : Prognostics and health management
SDP : Scaled dot product attention
Mamba : Linear-time sequence modeling with selective state spaces
FFT : Fast Fourier transform
ICFFT : Improved channel fast Fourier transform
CNorm : Cross-normalization
CNN : Convolutional neural network
RNN : Recurrent neural network
LSTM : Long short-term memory
MAE : Mean absolute error
RMSE : Root mean square error
x : Input data/input sequence
Q,K,V : Query, key, value vectors in attention
$h(t)$: Hidden state at time t
$y(t)$: Output at time t
A,B,C,D : State-space model matrices
t : Time step/time index
E_t : Error at time t
X : Multisensor fused data
N : Number of samples or sensors
α, β : Weights in scoring function

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